

Fronthaul Load Management in Open RAN

Uncertainty-aware, deadline-feasible budget allocation on a shared 7-2x fronthaul link

B.Tech. Project — mid-term evaluation

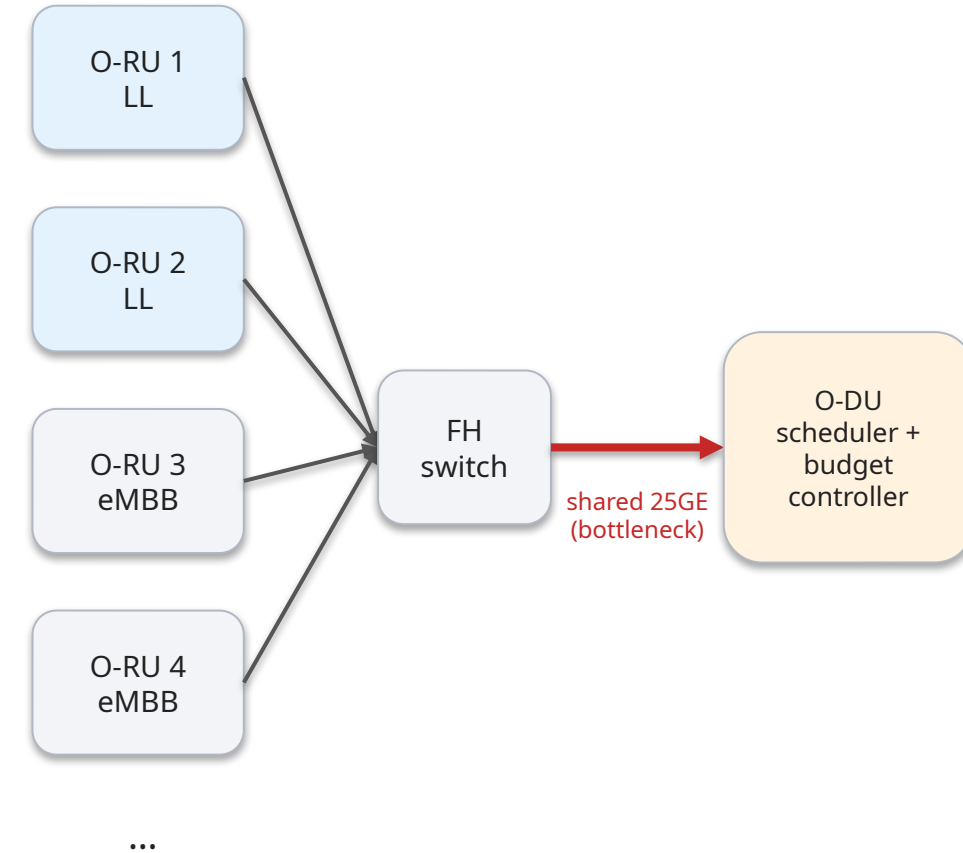
Student: <name> | Supervisor: <name> | Department of <dept>

Deliverables today: working simulator + 7 controllers + trained quantile forecaster + reproducible experiments (5 scenarios x 5 seeds) + tests + this deck, all generated from the repository.

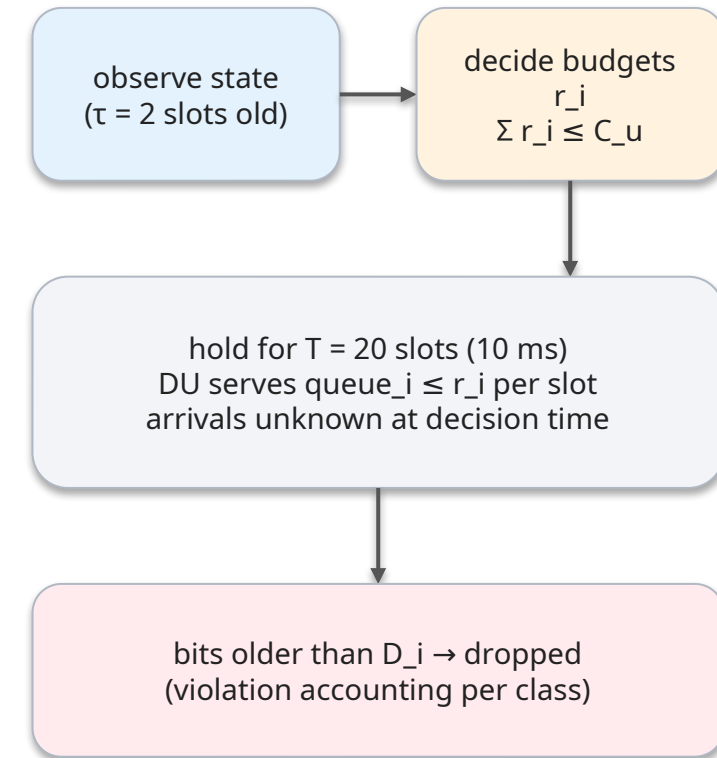
Why fronthaul load management?

1 Motivation

- O-RAN 7-2x split: O-RU does low-PHY; the fronthaul carries frequency-domain IQ per PRB, layer, symbol
 - Load-dependent: 3,387 FH bits per scheduled PRB-layer-slot (BFP-9, 8% eCPRI/Eth. overhead)
 - One 100 MHz / 4-layer cell peaks at 7.4 Gbit/s
- Shared aggregation links are over-subscribed on purpose (statistical multiplexing)
 - Default scenario: 8 cells, 25GE link, aggregate peak = 2.4x usable capacity
- When bursts coincide, some cell's IQ must be deferred or dropped; low-latency cells have ~2 ms of slack
- **Load management = deciding, every control interval, how much link each cell may use — before the traffic is known**
 - Bad decisions → deadline violations (lost IQ / HARQ failures) or wasted link capacity



- Time: NR slots of 0.5 ms (30 kHz SCS). Data: bits. Capacity: bits/slot
- Cells: 3 low-latency (deadline 4 slots = 2 ms, priority 3) + 5 eMBB (20 slots = 10 ms, priority 1)
- Demand mapping: FH bits = user bits / $SE_i(t) \times IQ$ bits per PRB-layer (SE differs per cell and drifts)
- Per-cell DU queue, FIFO, age-tracked; bits older than D_i are dropped = deadline violation
- Shared link: $\sum_i r_i \leq C_u = 0.97 \times 25$ Gbit/s (3% reserved for C/M/S-plane); $r_i \leq$ per-cell peak
- Controller: every $T = 20$ slots (10 ms) sets budgets r_i , fixed for the interval; sees state $\tau = 2$ slots old
- DU scheduler: serves each queue up to r_i per slot within the budget; link never queues (budgets enforce this)
- Why decide in advance? Budgets are a reservation held for T slots and telemetry is delayed $\rightarrow$ the next
 - interval's arrivals are unknown at decision time; the ~ 10 ms control interval matches real DU/switch reconfiguration



Sources: O-RAN WG4 CUS-Plane spec (7-2x sections, BFP, T2a windows); Larsen et al., IEEE COMST 2019 (load-dependent bit rate, $\sim 8\%$ Ethernet overhead); Pérez et al. 2018/2019 (packet-switched aggregation).

- **A. Fronthaul itself (primary sources)**

- O-RAN WG4 CUS-Plane spec (7-2x, BFP, T2a windows) • Larsen et al., IEEE COMST 2019: bit rate varies with load for splits $\leq 7-2$
- Pérez et al., JOCN 2018 / IEEE Access 2019 (packet-switched eCPRI aggregation) • Wang & Zhou, IEEE Commun. Lett. 2017 (multiplexing gain)
- Lagén et al., IEEE Commun. Mag. 2022: shared 7-2x link, reactive per-slot compression control — closest fronthaul work, no forecasting

- **B. Predictive / anticipatory capacity allocation**

- Bega et al., DeepCog (INFOCOM 2019 / JSAC 2020), AZTEC (INFOCOM 2020): DL capacity forecasts → slice allocation; fixed cost ratio
- Predictive DBA for PON fronthaul (Mikaeil et al. 2018; Zhang et al. 2019); Kavehmadavani et al., TWC 2023: LSTM point forecasts + FH constraint
- Typical treatment of uncertainty: a fixed safety margin on a point forecast; no per-cell deadline feasibility

- **C. Uncertainty-aware forecasting / conformal methods**

- Cohen et al., IEEE WCL 2023 (conformal URLLC pre-allocation, single stream) • Gibbs & Candès, NeurIPS 2021 (ACI)
- Kasuluru et al. 2024 [preprint]: probabilistic PRB-load forecasts, percentile chosen globally, not from a coupled allocation

- **Gap: no work turns a predictive DISTRIBUTION of deadline-feasible demand into per-cell FH budgets under one hard link constraint, and tests it against strong reactive + point-forecast baselines**

- Decision each epoch k : budgets $r \in \{0 \leq r_i \leq \text{cap}_i, \sum r_i \leq C_u\}$, held for slots $kT \dots (k+1)T-1$
- **Primary metric (fixed before tuning): priority-weighted deadline-violation ratio**
 - $V_w = \sum_i p_i \cdot \text{violated_bits}_i / \sum_i p_i \cdot \text{arrived_bits}_i$ ($p_{LL} = 3, p_{eMBB} = 1$)
 - Supporting: per-class violation ratio, link utilisation, mean/p99 DU queueing delay, Jain fairness of service ratio, decision time
- **Deadline-feasible rate $r^*(a, D)$: smallest constant rate that serves arrivals $a_1..a_H$ with no bit waiting $> D$ slots**
 - $r^*(a, D) = \max \text{ over windows } [s, e] \text{ of } \sum_{t=s..e} a_t / (e - s + 1 + D)$ — network-calculus bound, exact for FIFO
 - Demand of a cell for the next interval = backlog-clearing rate + r^* of the unseen arrivals
- Why forecasting matters: r^* of the NEXT interval is not observable; LL $r^* \approx$ peak 4-slot burst rate (hard to predict)
- Why not point forecasts: r^* is right-skewed (bursts) $\rightarrow$ the median under-reserves, a fixed margin over-reserves everywhere

Baseline controllers (implemented; tuned on validation seeds)

5 Baselines

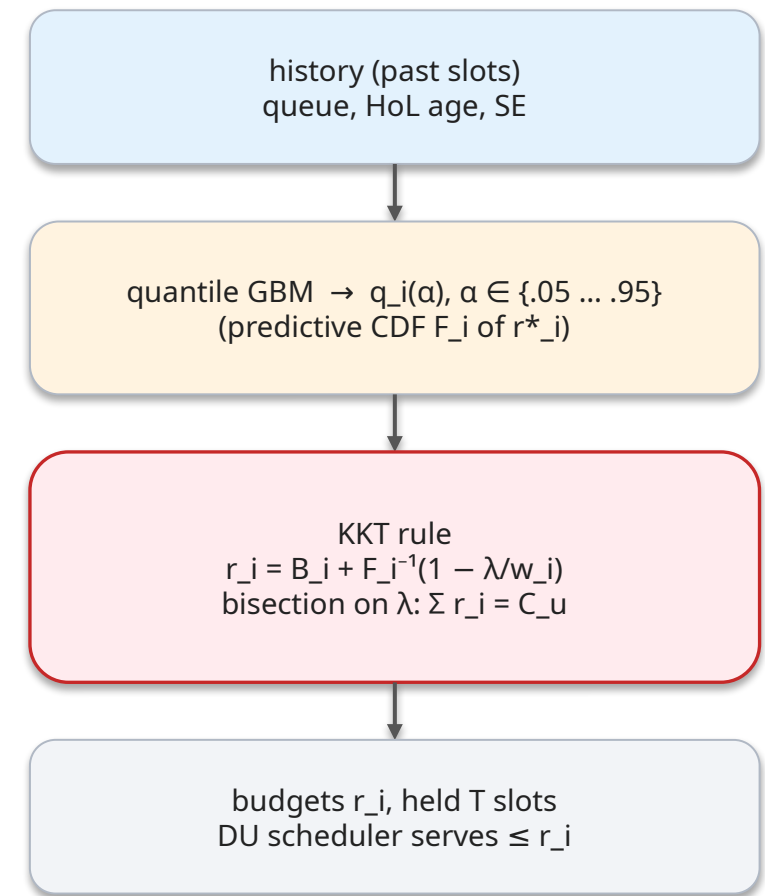
Controller	Budget rule	Information	Tuned
Static equal share	C_u / N per cell (caps redistributed)	none	—
Reactive proportional	$\text{backlog}/T + (1+m) \cdot \text{EWMA}$; proportional split; leftover redistributed	history	m , EWMA window
Deadline-aware reactive (strong)	$\text{backlog-slack rate} + (1+m) \cdot r^*(\text{last horizon})$; priority weights; water-fill	history, D_i , p_i	m , #horizons
Point forecast + water-fill (ablation)	same as above with GBM MEDIAN forecast of r^*	history, D_i , p_i , model	m
Oracle water-fill (reference)	true r^* of the next horizon; same water-fill as above (perfect point forecast, not a bound on other rules)	future arrivals	—

- **Validation tuning (seeds 2000-2003, scenarios moderate/high; test seeds untouched):**

- $\text{reactive_prop} \rightarrow \{\text{'margin': 0.0, 'ewma_window': 20}\}$ | $\text{queue_aware} \rightarrow \{\text{'margin': 0.0, 'num_horizons': 1}\}$ | $\text{point_forecast} \rightarrow \{\text{'margin': 0.0}\}$ | $\text{proposed_cal} \rightarrow \{\text{'gamma': 0.005}\}$
- Finding: every safety margin > 0 was worse on validation — inflating a point estimate hurts when the shortage is split proportionally (over-claimers get more)

- All controllers: identical traces, identical delayed observations, unused capacity always redistributed

- **1. Forecaster: $K = 6$ quantiles (5, 25, 50, 75, 90, 95%) of r^*_i for the next horizon**
 - One HistGradientBoostingRegressor(loss='quantile') per level; 28 features from past slots only (lag means, EWMA, last slots, past r^* , SE, deadline, cell id)
 - Target normalised by the cell's fixed peak $\rightarrow$ no statistics fitted on data; trained on TRAIN seeds only
- **2. Allocator: separable concave program under the forecast distribution F_i**
 - $\max \sum_i w_i \cdot E[\min(B_i + r^*_i, r_i)]$ s.t. $\sum r_i \leq C_u$, $0 \leq r_i \leq \text{cap}_i$
 - KKT: $r_i(\lambda) = \text{clip}(B_i + F_i^{-1}(1 - \lambda / w_i), 0, \text{cap}_i)$, λ from bisection on $\sum r_i(\lambda) = C_u$ ($\lambda = 0$ if slack $\rightarrow$ headroom spread)
 - Meaning: every cell is cut at the same weighted tail probability $w_i \cdot P(\text{marginal bit needed}) = \lambda$
 - $\rightarrow$ wide forecast or high weight $\Rightarrow$ larger margin; confident forecast $\Rightarrow \approx$ median; backlog funded first ($F_i(x < 0) = 0$)
- Optional: per-cell ACI offset δ_i on the quantile level (Gibbs & Candès 2021) as an online fallback — tested as ablation
- Cost: $O(N \cdot K + N \cdot 60)$ per epoch; feasibility by construction (Prop. 1-3 in docs/fhlm/method.md)



*Adapted: quantile GBM, ACI, water-filling.
Candidate contribution: r^* as forecast target + KKT rule over the forecast CDF under the link constraint, evaluated vs strong baselines.*

config.py	units, 7-2x FH constants, cells, link	traffic.py	synthetic arrivals: Gamma + Pareto ON/OFF + regimes + SE drift
demand.py	deadline-feasible rate r^* , backlog slack rate	simulator.py	slot loop, FIFO age queues, drops, metrics
controllers.py	5 baselines incl. oracle, water-fill	forecast.py	features, quantile GBM, window quantiles, CompiledGBM
proposed.py	KKT allocator + ACI offsets	scenarios.py	train/val/test seeds, 5 scenarios
train.py	train on TRAIN seeds, validate on VAL	run_experiments.py	tune / main / sweep / demo; raw JSON per run
make_figures.py	all figures from CSV	tests/test_fhlm.py	13 tests (capacity, queues, accounting, leakage, KKT ...)

- Pipeline: traffic → simulator ↔ controller (observation τ old → budgets) → metrics/raw JSON → figures → slides
- Data flow for ML: TRAIN seeds → features/targets → 6 GBMs (~12 s CPU) → VAL coverage check → frozen model → TEST
- Python 3, numpy/pandas/scikit-learn/matplotlib only; full reproduction ≈ 6 min on a 4-core laptop

Experimental setup (synthetic traffic, labelled as such)

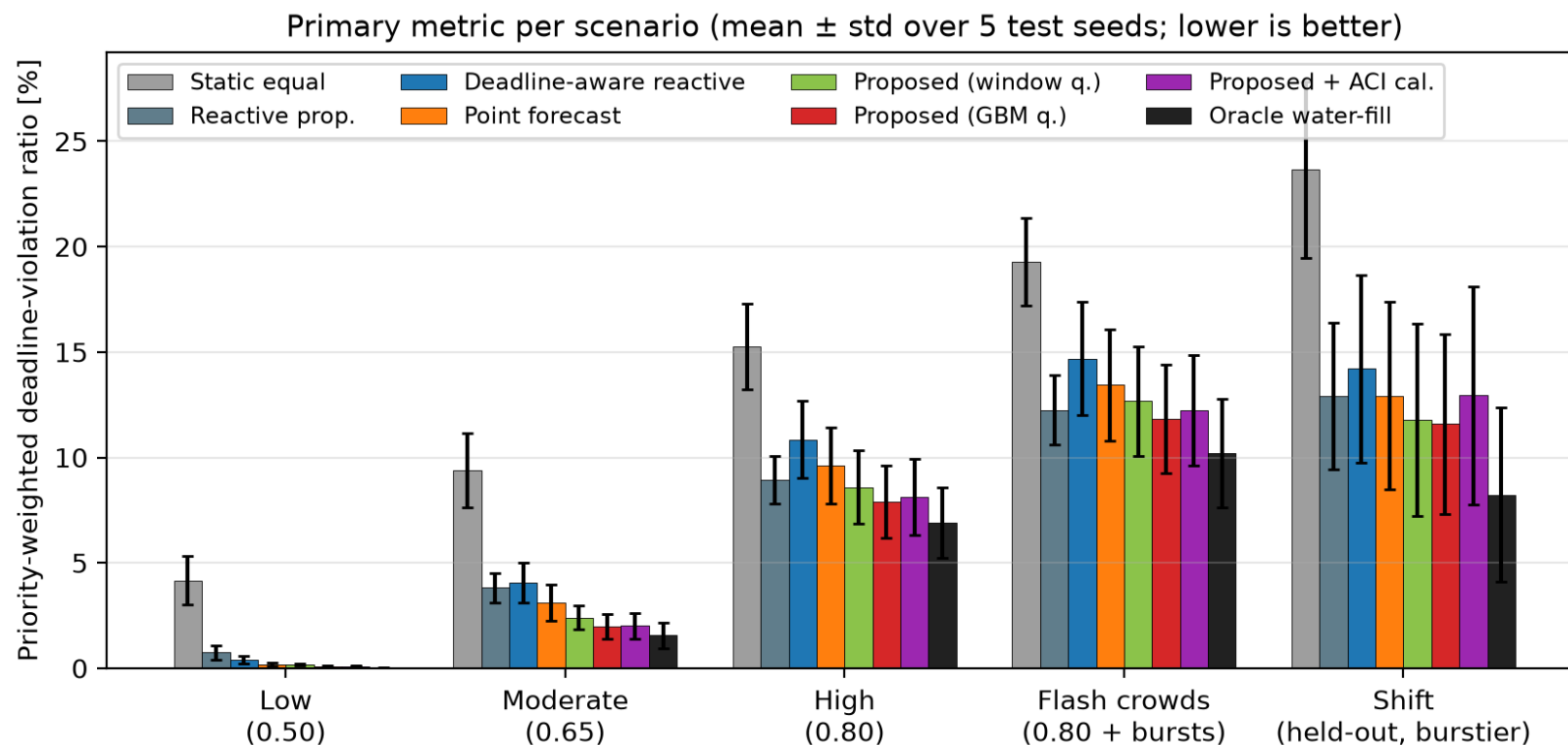
8 Experiments

Scenario	Mean FH load / C_u	Traffic family	Used for tuning?
low	0.50	nominal	no
moderate	0.65	nominal	yes (validation seeds)
high	0.80	nominal	yes (validation seeds)
overload_bursts	0.80	flash crowds: $\times 1.8$ regimes, 15% of the time (offered load $> C_u$)	no
shift (held-out)	0.80	ON rate $\times 1.5$, ON duration $\times 1.5$, Pareto $\alpha_{on} = 1.2$ — never in training/tuning	no

- Traffic (synthetic): Gamma smooth part + Pareto ON/OFF bursts ($\alpha_{on} 1.5$) + log-normal regimes (~ 1 s) + SE drift
- Seeds: train 1000-1015 (loads 0.5-0.95) | validation 2000-2003 | test 3000-3004 — disjoint
- Per run: 20,000 slots = 10 s after 1,000-slot warm-up; $T = 20$ slots, $\tau = 2$ slots; 8 methods $\times$ 5 scenarios $\times$ 5 seeds = 200 runs
- Identical traces and identical delayed observations for every controller; forecaster frozen before test
- Note: absolute violation ratios are high by design (heavy-tailed bursts at 0.8 mean load); relative, paired differences are the evidence

Results: primary metric per scenario (5 test seeds, mean $\pm$ std)

9 Results

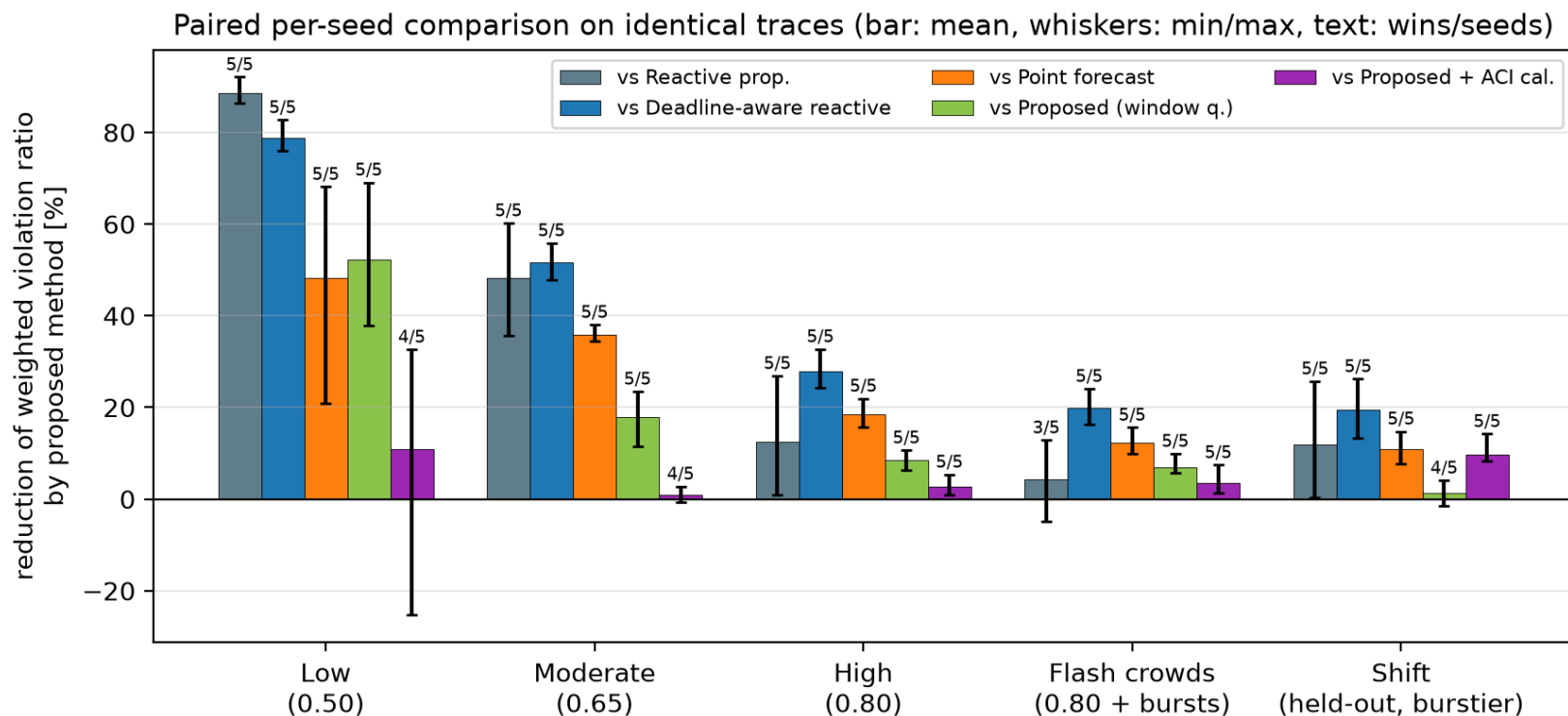


• Proposed = best implementable method in 5/5 scenarios

- vs point forecast (same model, median only): –18% (high), –10% (shift)
- vs deadline-aware reactive: –27% (high), –18% (shift)
- vs reactive proportional: –12% (high) but only –3% under flash crowds (3/5 seeds) → tie
- Gap to oracle at high load: 7.9% vs 6.9%
- Std across seeds is large (burst timing) → paired per-seed comparison next

Results: paired per-seed comparison on identical traces

9 Results



- **Wins 5/5 seeds vs point forecast and vs deadline-aware reactive in every scenario**

- vs reactive proportional: 5/5 in four scenarios; 3/5 under flash crowds (no claim there)

- **ML ablation (same rule, window quantiles instead of GBM):**

- GBM adds 2–51% ; smallest under held-out shift (2%) → the rule, not the model, carries most of the gain

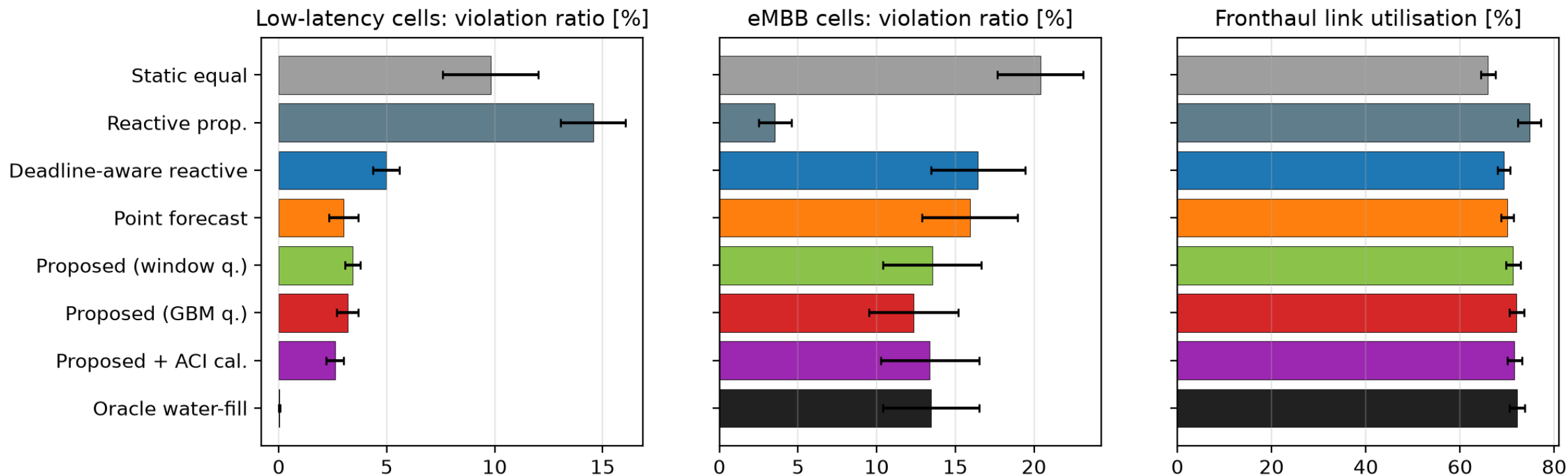
- ACI calibration: no gain in-distribution, harmful under shift (over-reserves LL, starves eMBB) → negative result

- No significance test claimed with 5 seeds; win counts and min/max are reported instead

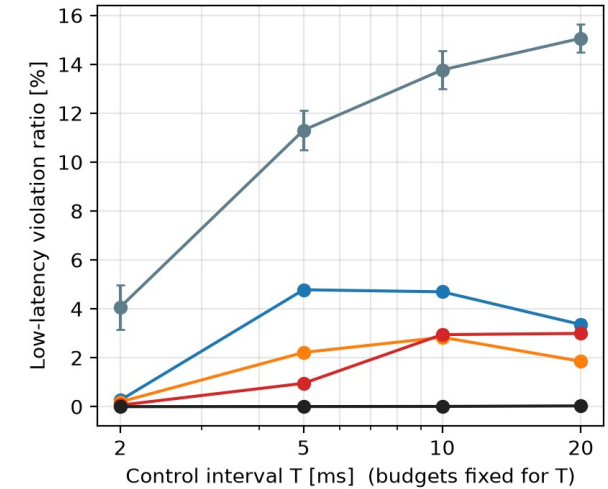
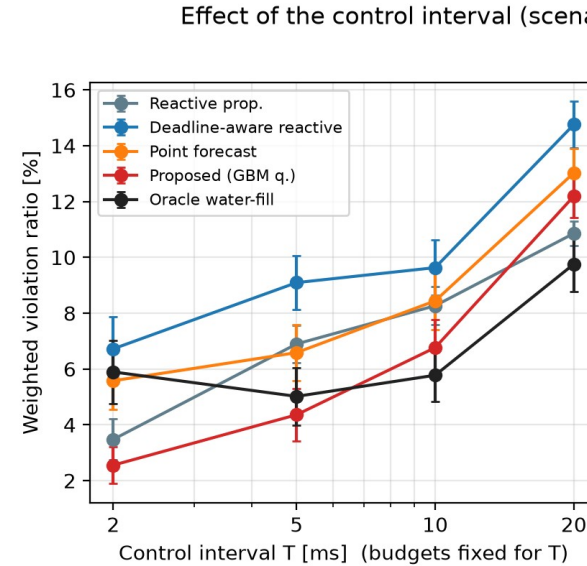
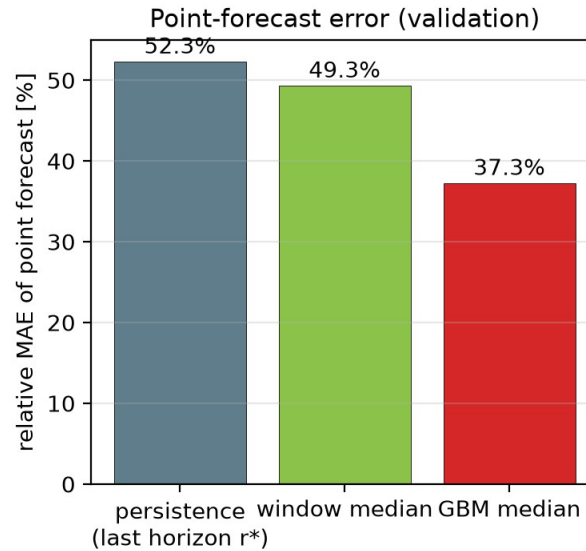
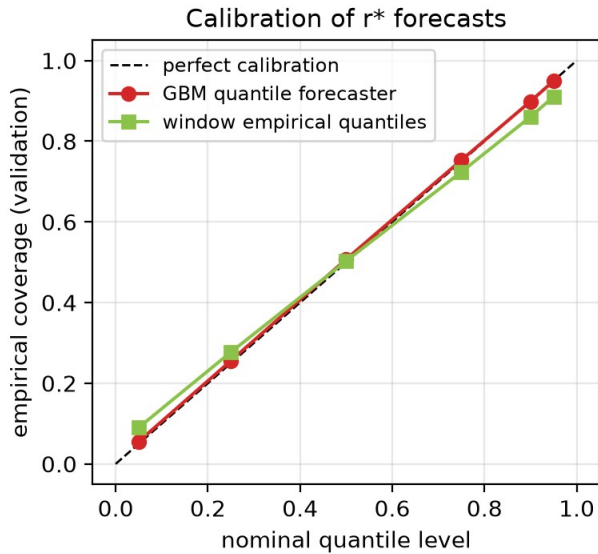
Where the gain comes from: per-class trade-off (high load)

9 Results

Scenario 'high': per-class trade-off (mean $\pm$ std over seeds)



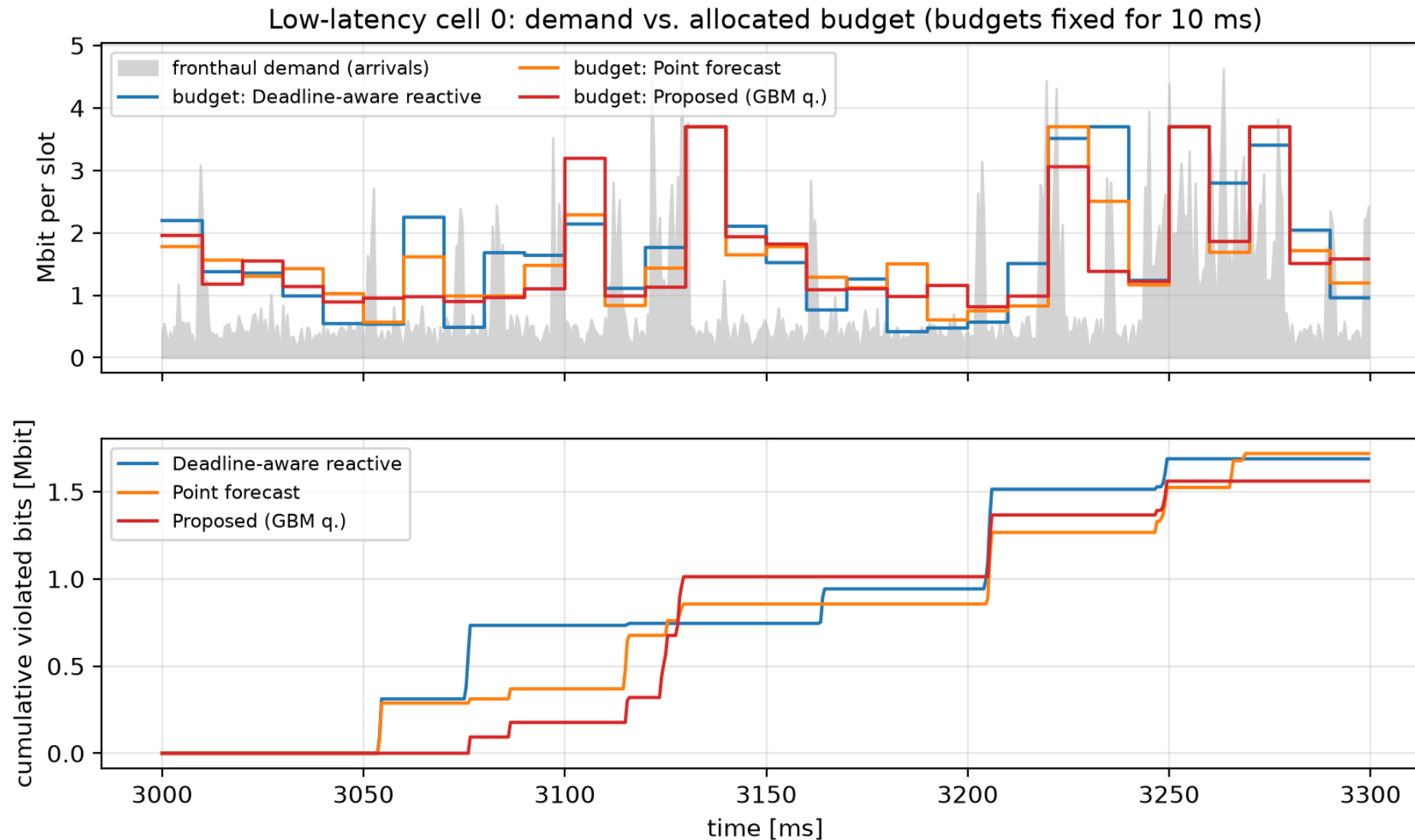
- Reactive proportional: LL 14.6% vs proposed 3.2% — deadline-unaware controllers sacrifice LL
- Within the deadline-aware family, proposed has the lowest eMBB loss (12.3% vs 15.9% point forecast) and highest utilisation (72.0%)
- Oracle still loses 13.4% eMBB at 0.80 load: 10-ms fixed budgets cannot track heavy-tailed bursts → control interval matters



- Validation coverage at nominal 5/50/95%: 5.6% / 50.7% / 94.9% ; median rel. MAE 37% vs persistence 52% , window 49%
- **Interval sweep (scenario high, 3 seeds), reduction by proposed vs deadline-aware reactive / vs reactive proportional / vs oracle water-fill:**
 - T=2 ms: +62% / +27% / +57% | T=5 ms: +52% / +37% / +13% | T=10 ms: +30% / +18% / -17% | T=20 ms: +17% / -12% / -25%
- At T = 2 ms the KKT rule even beats the oracle water-fill (perfect point forecast, proportional shortage split) → the allocation RULE, not only the forecast, drives the gain
- At T = 20 ms (budgets held 5× the LL deadline) the deadline-aware family over-reserves the LL peak rate and reactive proportional wins on the weighted metric → known failure regime; a bound on how long a reservation should be held

One trace: budgets vs. demand of a low-latency cell (300 ms)

9 Results



- Synthetic traffic only (Gamma + Pareto ON/OFF + regimes); no real O-RU traces → no real-world validation claimed
- Gains are relative to interval-level controllers; the oracle itself is interval-level — a slot-level optimum is lower
- Not a universal win: tie with reactive proportional under flash crowds; GBM adds $\approx 0\text{-}2\%$ over window quantiles under held-out shift
- ACI online calibration did not help (negative result); the robust fallback is currently the window-quantile variant
- 5 test seeds → win counts and min/max reported, no significance test
- Simplified link: budget-enforced pipe, no switch packet queueing, one traffic class per cell, no HARQ/RLC feedback
- Decision time ≈ 3.4 ms per 10-ms epoch in Python (vectorised tree evaluation); reactive baselines ≈ 0.1 ms
- Propositions in docs are feasibility / equal-tail-probability / backlog-first — no stability or optimality-in-the-loop claims

Implemented and validated

- Slot-level 7-2x shared-link simulator with deadlines, priorities, telemetry delay, accounting (13 tests)
- Synthetic heavy-tailed traffic generator with regimes, flash crowds, SE drift
- 5 baselines (static, proportional, deadline-aware reactive, point forecast, oracle), tuned on validation seeds
- Quantile-GBM forecaster of r^* (calibrated on validation), vectorised inference
- KKT allocation rule with feasibility proof; ACI variant; window-quantile variant
- 200-run main comparison, paired analysis, interval sweep, demo, all figures, this deck — reproducible

Planned (not yet done)

- Real or public traffic traces (e.g. per-cell PRB utilisation datasets) → re-validate the forecaster
- Skill-weighted blend of GBM and window quantiles as a principled fallback (replace ACI)
- Second control lever: per-cell BFP compression / modulation compression under the same rule
- Switch-level packet queue and T2a window model instead of a budget-enforced pipe
- Ablation on the target: r^* vs interval volume; more seeds; sensitivity to τ and priorities
- Cost model: does the 3 ms Python decision fit a real DU? (C++/numba port, or longer epochs)

Backup: primary metric [%], mean $\pm$ std over 5 test seeds

Backup

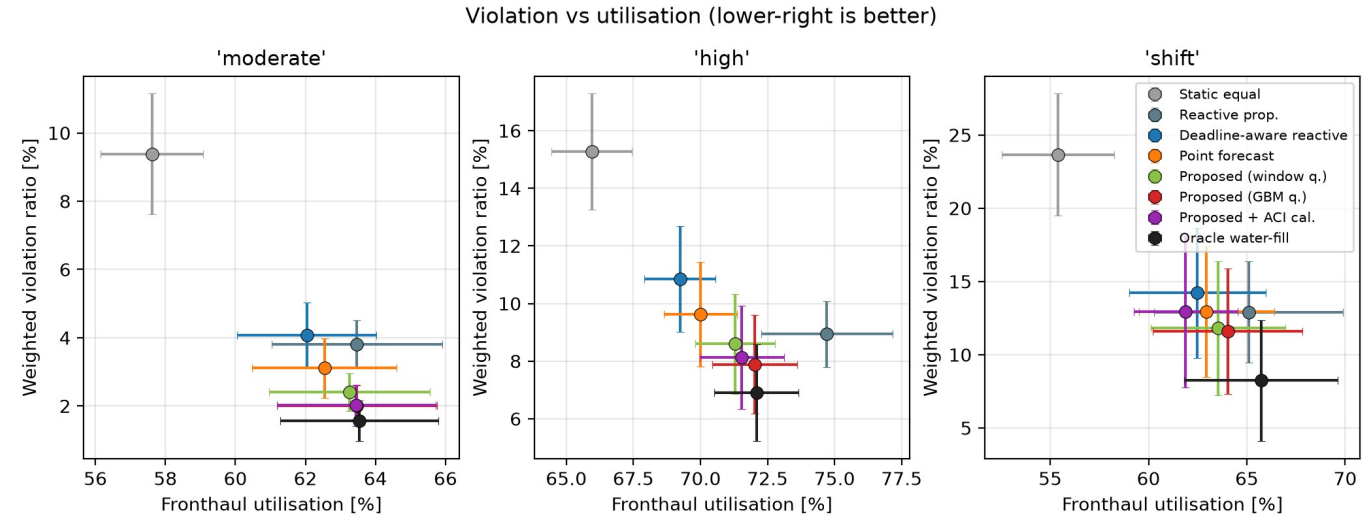
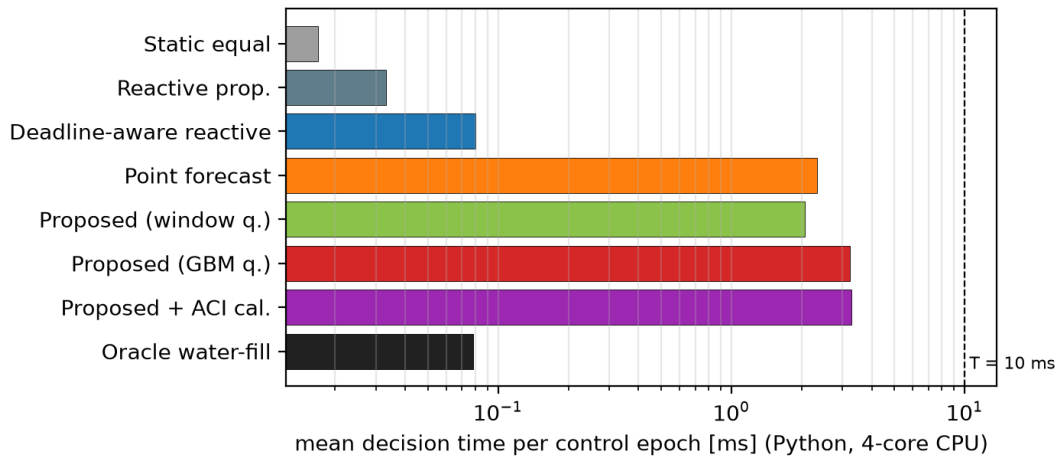
Method	Low 0.50	Moderate 0.65	High 0.80	Flash crowds	Shift (held-out)
Static equal share	4.16 $\pm$ 1.15	9.38 $\pm$ 1.77	15.27 $\pm$ 2.02	19.30 $\pm$ 2.07	23.65 $\pm$ 4.18
Reactive proportional (EWMA)	0.75 $\pm$ 0.34	3.81 $\pm$ 0.71	8.94 $\pm$ 1.15	12.25 $\pm$ 1.65	12.92 $\pm$ 3.47
Deadline-aware reactive (persistence r^*)	0.39 $\pm$ 0.18	4.07 $\pm$ 0.95	10.86 $\pm$ 1.84	14.70 $\pm$ 2.67	14.21 $\pm$ 4.44
Point forecast (GBM median)	0.18 $\pm$ 0.10	3.11 $\pm$ 0.87	9.63 $\pm$ 1.82	13.44 $\pm$ 2.65	12.93 $\pm$ 4.45
Proposed rule, window quantiles (no ML)	0.17 $\pm$ 0.04	2.41 $\pm$ 0.56	8.60 $\pm$ 1.74	12.67 $\pm$ 2.59	11.80 $\pm$ 4.56
Proposed (GBM quantiles + KKT)	0.08 $\pm$ 0.04	2.00 $\pm$ 0.59	7.89 $\pm$ 1.71	11.82 $\pm$ 2.56	11.59 $\pm$ 4.27
Proposed + ACI calibration	0.09 $\pm$ 0.02	2.02 $\pm$ 0.61	8.12 $\pm$ 1.80	12.24 $\pm$ 2.62	12.95 $\pm$ 5.17
Oracle water-fill (true r^* , reference)	0.01 $\pm$ 0.01	1.56 $\pm$ 0.60	6.90 $\pm$ 1.68	10.21 $\pm$ 2.57	8.23 $\pm$ 4.14

Source: results/fhlm/main_runs.csv (200 runs). Paired per-seed statistics: results/fhlm/main_paired.csv.

- Objective: $J(r) = \sum_i w_i E_{\{F_i\}}[\min(B_i + X_i, r_i)]$, $X_i \sim F_i$ (forecast CDF of r_i^*)
- $d/dr_i E[\min(B_i + X_i, r_i)] = P(B_i + X_i > r_i) = 1 - F_i(r_i - B_i) \rightarrow J$ concave, separable
- Lagrangian: $L = J - \lambda(\sum r_i - C_u) + \sum \mu_i r_i - \sum v_i (r_i - \text{cap}_i)$
- Stationarity for an interior r_i : $w_i (1 - F_i(r_i - B_i)) = \lambda \Rightarrow r_i = B_i + F_i^{-1}(1 - \lambda / w_i)$
- Clip at 0 and cap_i ($\mu_i, v_i \geq 0$); $\sum_i r_i(\lambda)$ is non-increasing in $\lambda \Rightarrow$ unique λ^* by bisection (60 steps)
- Prop. 1 (feasibility): output always satisfies caps and $\sum r_i \leq C_u$ — verified by `test_kkt_allocation_properties`
- Prop. 2 (equal weighted tail): at the optimum all interior cells share $w_i \cdot P(\text{marginal bit needed}) = \lambda^*$
- Prop. 3 (backlog first): $F_i(x) = 0$ for $x < 0 \Rightarrow$ marginal value below B_i is $w_i \Rightarrow$ backlog of any cell with $\lambda^* < w_i$ is fully funded
- Degenerate F_i (point forecast) $\Rightarrow$ priority-ordered fill = the point-forecast water-fill baseline
- F_i^{-1} : piecewise-linear through the 6 forecast quantiles, linear extrapolation above 0.95; monotone by sorting

Backup: decision overhead and utilisation trade-off

Backup



Decision times are Python on a 4-core CPU inside the simulator (feature construction + 1,800 trees via vectorised traversal + bisection). The rule itself is < 0.1 ms.

- O-RAN WG4, Control, User and Synchronization Plane Specification (O-RAN.WG4.CUS.0) [standard] — 7-2x, BFP compression (Annex A.1), T2a windows. 3GPP TR 38.801 [standard]; IEEE 802.1CM-2018 [standard].
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- W. Willinger et al., IEEE/ACM ToN 1997 (heavy-tailed ON/OFF sources); Hadley & Whitin 1963 (constrained multi-item newsvendor); Le Boudec & Thiran, Network Calculus, 2001. Full list: docs/fhlm/literature_review.md